

Advanced Data Management

Logical Formalization of Information Integration Systems

Domenico Fabio Savo

*Corso di laurea magistrale
INGEGNERIA INFORMATICA
Dipartimento di Ingegneria Gestionale, dell'Informazione e della Produzione
University of Bergamo*

Outline of the course

1. Introduction to propositional logic
 2. Introduction to First-order Logic
 3. Relational Calculus
 4. Information Integration **S**ystems (IIS)
 5. **Logical formalization of IISs**
 6. Mapping between **Global Schema** e **Data Sources**
 7. Incomplete information databases
 8. Query answering over IISs
 9. **O**ntology-**B**ased **D**ata **M**anagement (OBDM)
 10. Description Logic Ontologies
 11. Query answering in ontologies
 12. Query answering in OBDM
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Logical Formalization

Formalization of Information Integration Systems

- The architecture of an Information Integration Systems (I.I.S.) can be defined using **First-Order Logic**
 - However, we need to make some simplifying assumptions
- Data sources are represented as one single relational database
 1. No syntactic heterogeneity between sources
 2. A single logical theory (**source schema**) describes valid data sources
 3. Source schemas may contain constraints (FOL sentences)
- The result of the integration is a relational database.
 1. No syntactic heterogeneity between sources and the result of integration
 2. A single logical theory (**global schema**) describes the result of the integration
 3. Global schemas may contain constraints (FOL sentences)

Information Integration Specification - Syntax

An information integration specification is a triple $J = \langle G, M, S \rangle$, where:

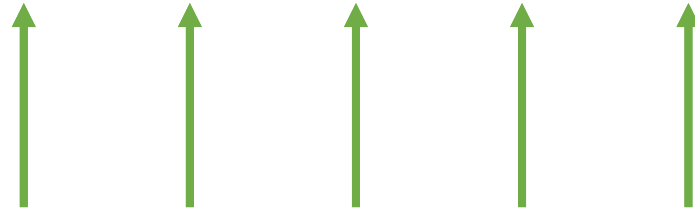
- **G** is the **global schema**
 - A relational database schema that describes how data resulting from the integration process should look like
 - A database **B** for **G** will be called **global database**, i.e., we assume that $B \models G$
- **S** is the **source schema**
 - A relational database schema describing the data sources such that $G \cap S = \emptyset$
 - A database **D** for **S** will be called **source database**, i.e., we assume that $D \models S$
- **M** is the **mapping** between **source schema S** and **global schema G**
 - A set of mapping assertions $m = \langle q_S, q_G \rangle$, where q_S is a FOL query over **S** and q_G is a FOL query over **G**. q_S and q_G have the same arity!

Information Integration Specification

Global Schema G



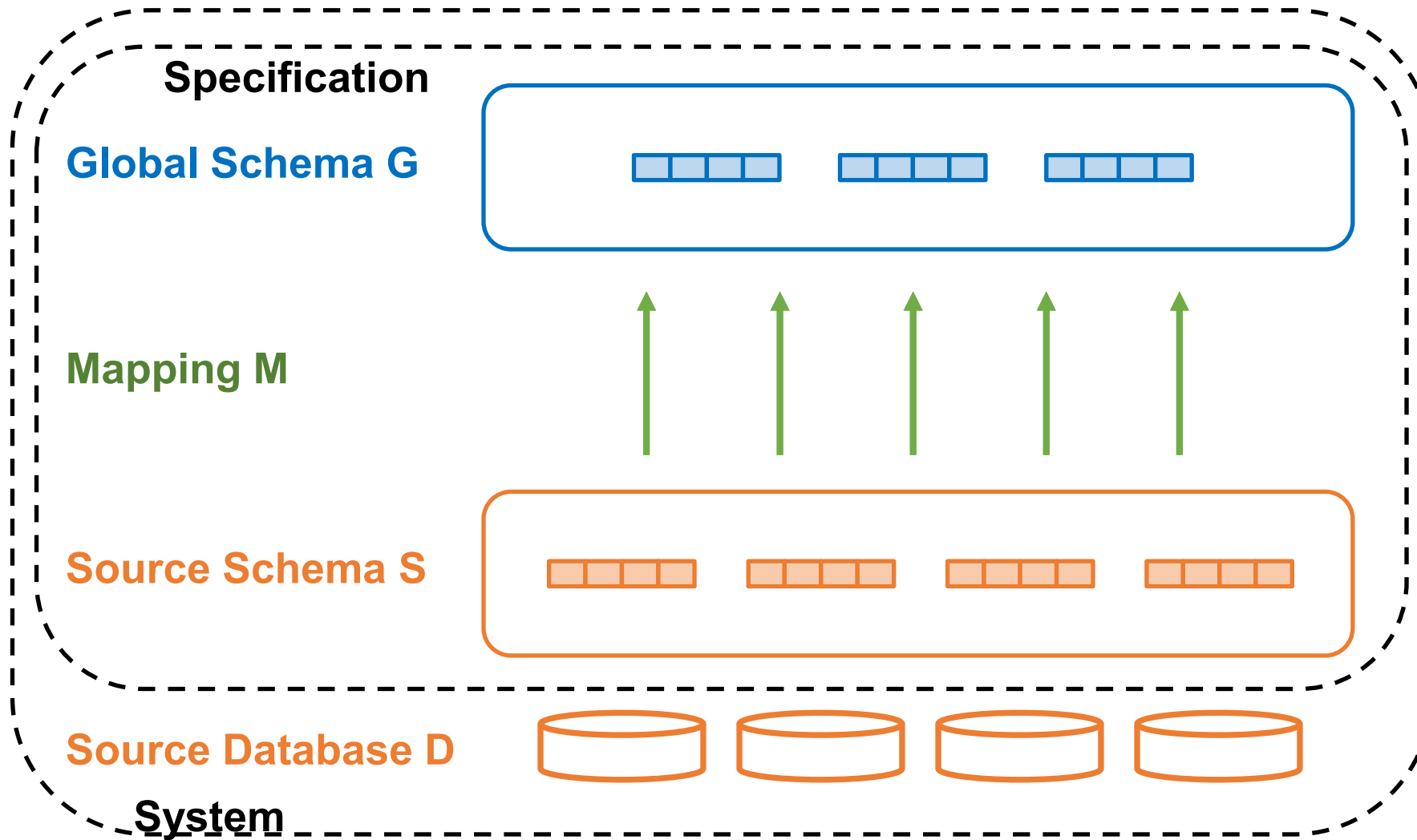
Mapping M



Source Schema S



Information Integration System - Syntax



An information integration **system** is a pair $\langle \mathbf{J}, \mathbf{D} \rangle$, where:

- $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration **specification**
- $\mathbf{D}$ is a **source database** for $\mathbf{S}$, i.e., $\mathbf{D} \models \mathbf{S}$

Information Integration System - Semantics

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

- A global database $\mathbf{B}$ **satisfies** a mapping assertion $\mathbf{m} = \langle q_S, q_G \rangle$ in $\mathbf{M}$ with respect to $\mathbf{D}$, denoted by $(\mathbf{B}, \mathbf{D}) \models \mathbf{m}$, if $q_S(\mathbf{D}) \subseteq q_G(\mathbf{B})$
(*sound source assumption*)
- A global database $\mathbf{B}$ **satisfies** the mapping $\mathbf{M}$ with respect to $\mathbf{D}$, denoted by $(\mathbf{B}, \mathbf{D}) \models \mathbf{M}$, if $(\mathbf{B}, \mathbf{D}) \models \mathbf{m}$ for each $\mathbf{m} \in \mathbf{M}$

Intuitively, the semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$ is the set of **global databases**, i.e., is a database for $\mathbf{G}$, that satisfy the **mapping** w.r.t. to the **source database**.
Formally:

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid \mathbf{B} \text{ is a global database s.t. } (\mathbf{B}, \mathbf{D}) \models \mathbf{M} \}$$

Source Schema and Database

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Purchases

ID	Customer Name	Product
O001	Amber	P001
O002	Dylan	P001

Global Schema

Person

Id	Name
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Example of Information Integration System

Assume $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ with:

- $\mathbf{G} = \{Person_{/2}\}$
- $\mathbf{S} = \{Purchases_{/3}, Customers_{/3}\}$
- $\mathbf{M} = \{m_1, m_2\}$ where:
 - $m_1 = \langle \{(n) \mid \exists id. \exists p. Purchases(id, n, p)\}, \{(n) \mid \exists i. Person(i, n)\} \rangle$
 - $m_2 = \langle \{(n) \mid \exists d. \exists r. Customers(n, d, r)\}, \{(n) \mid \exists i. Person(i, n)\} \rangle$

Assume a **source database D** as defined previously

See exercises in the next slides!

Exercise 1

Provide a global database over $\mathbf{G} = \{Person_{/2}\}$ that belongs to $\text{sem}(\mathbf{J}, \mathbf{D})$

$\mathbf{M} = \{m_1, m_2\}$ where:

- $m_1 = \langle \{(n) \mid \exists id. \exists p. \text{Purchases}(id, n, p)\}, \{(n) \mid \exists i. \text{Person}(i, n)\} \rangle$
- $m_2 = \langle \{(n) \mid \exists d. \exists r. \text{Customers}(n, d, r)\}, \{(n) \mid \exists i. \text{Person}(i, n)\} \rangle$

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Purchases

ID	Customer Name	Product
O001	Amber	P001
O002	Dylan	P001

Exercise 1 - Solution

Provide a global database over $\mathbf{G} = \{Person_{/2}\}$ that belongs to $\text{sem}(\mathbf{J}, \mathbf{D})$

Person

Id	Name
1	Amber
2	Rose
3	Dylan

Exercise 2

Which of the following global databases belong to $\text{sem}(\mathbf{J}, \mathbf{D})$? Why?

B1

Id	Name
1	Amber
2	Rose
3	Dylan
2	Dylan

B2

Id	Name
1	Amber
2	Jonny
3	Dylan

B3

Id	Name
a	Amber
b	Rose
c	Dylan
d	Jonny

Exercise 2 - Solution

Which of the following global databases belong to $\text{sem}(\mathbf{J}, \mathbf{D})$? Why?

B1

Id	Name
1	Amber
2	Rose
3	Dylan
2	Dylan

B2

Id	Name
1	Amber
2	Jonny
3	Dylan

B3

Id	Name
a	Amber
b	Rose
c	Dylan
d	Jonny

B1: Yes!

B2: No! Rose is missing

B3: Yes!

Querying an Information Integration System

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

Querying $\langle \mathbf{J}, \mathbf{D} \rangle$ simply means posing queries over the **global schema** $\mathbf{G}$ of the information integration specification $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$

Intuitively, the **certain answers** for a FOL query q w.r.t. $\langle \mathbf{J}, \mathbf{D} \rangle$ are those tuples of constants that occur in the answers for q over **every possible global database** in the semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$. Formally:

$$\text{cert}(q, \mathbf{J}, \mathbf{D}) = \{(c_1, \dots, c_n) \mid (c_1, \dots, c_n) \in q(\mathbf{B}) \text{ for each } \mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})\}$$

Exercise 3

Consider the previous information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, and consider the following queries:

$$q_1 = \{ () \mid \exists n. \exists i. \text{Person}(i, n) \}$$

$$q_2 = \{ (n) \mid \exists i. \text{Person}(i, n) \}$$

$$q_3 = \{ (i) \mid \exists n. \text{Person}(i, n) \}$$

$$q_4 = \{ (i, n) \mid \text{Person}(i, n) \}$$

Compute $\text{cert}(q_1, \mathbf{J}, \mathbf{D})$, $\text{cert}(q_2, \mathbf{J}, \mathbf{D})$, $\text{cert}(q_3, \mathbf{J}, \mathbf{D})$, and $\text{cert}(q_4, \mathbf{J}, \mathbf{D})$.

Exercise 3 - Solution

Consider the previous information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, and consider the following queries:

$$q_1 = \{ () \mid \exists n. \exists i. \text{Person}(i, n) \}. \text{cert}(q_1, \mathbf{J}, \mathbf{D}) = \{ () \} \text{ (true)}$$

$$q_2 = \{ (n) \mid \exists i. \text{Person}(i, n) \}. \text{cert}(q_2, \mathbf{J}, \mathbf{D}) = \{ (\text{Amber}), (\text{Rose}), (\text{Dylan}) \}$$

$$q_3 = \{ (i) \mid \exists n. \text{Person}(i, n) \}. \text{cert}(q_3, \mathbf{J}, \mathbf{D}) = \emptyset$$

$$q_4 = \{ (i, n) \mid \text{Person}(i, n) \}. \text{cert}(q_4, \mathbf{J}, \mathbf{D}) = \emptyset$$

For q_1 and q_2 , note that Amber, Rose, and Dylan must occur in all global databases $\mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})$

For q_3 and q_4 , consider the global databases $\mathbf{B1} \in \text{sem}(\mathbf{J}, \mathbf{D})$ and $\mathbf{B3} \in \text{sem}(\mathbf{J}, \mathbf{D})$ given before. Note that $q_i(\mathbf{B1}) \cap q_i(\mathbf{B3}) = \emptyset$ for both $i=3$ and $i=4$

Complexity of Query Answering

Complexity of Query Answering

- **Thm:** Let $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ be an information integration specification, $\mathbf{D}$ be a **source database** for $\mathbf{S}$, and $\mathbf{q}$ be a **FOL query** over $\mathbf{G}$. Computing $\text{cert}(\mathbf{q}, \mathbf{J}, \mathbf{D})$ is **undecidable**.
- This holds even if $\mathbf{J}$ has no mappings and no axioms in the schemas

Proof: Let $\mathbf{J} = \langle \emptyset, \emptyset, \emptyset \rangle$. Then, $\text{sem}^D(\mathbf{J})$ contains every interpretation I for the alphabet of $\mathbf{G}$. Therefore, $\text{cert}(q, J, D) \neq \emptyset$ if and only if $\mathbf{q}$ is defined by a *tautological* FOL formula. Undecidability follows from ***undecidability of FOL Tautology*** (checking whether a FOL formula is a tautology)

We will study well behaved languages in the following lectures.

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.
